Diego **Zamboni**

diego@zzamboni.org | zzamboni.org | Zurich, CH | zzamboni |
zzamboni | zzamboni | zzamboni.org

CISO • Organizational Leader • Security Expert • Computer Scientist

Senior computer scientist, security expert and organizational leader with 30 years of experience across security governance, cloud security architecture, research and engineering. I combine strategic leadership, technical depth and clear communication to help organizations design secure systems, scale teams and deliver measurable outcomes.

Professional Highlights

Management and leadership, IT security, cloud computing

- Chief Information Security Officer for Governance at Avaloq, defining and managing Avaloq's global ISO27001-certified Information Security Management System
- Managed security architecture at the Stellantis *Virtual Engineering Workbench* project. Worked with Stellantis CISO and business stakeholders to define governance, establish security best practices and drive risk analysis, threat modeling and mitigation.
- Established scalable and durable mechanisms to enable partners to work securely in the Volkswagen Digital Production Platform (DPP) program.
- Managed security architecture, risk management, data governance and compliance (ISO27001, ISAE3402/3000, etc.) for Swisscom's Cloud platforms.
- Established and led the Swisscom IT Clouds security community of practice.
- Established and led the *Health and State Management* team at Swisscom to design, implement and operate a framework for scalable monitoring, logging and alerting for Swisscom's Cloud platforms.
- Established and led the first computer security organization at UNAM, which has grown into the university's Information Security Coordination (UNAM-CERT).
- Managed IT security customer relationships at HP Enterprise Services, including overseeing the activities of operational and engineering teams, risk and compliance management, requirements discussion and reporting.

Research, architecture and design

- Designed the *Orchard* monitoring framework for Swisscom's *Application Cloud* platform, and led the team that implemented it and brought it into production.
- Designed and implemented the *Billy Goat* malware capture and analysis system at IBM.

Experience

Avaloq — *Switzerland*
CISO Governance — *Jan 2024 - Present*

Lead Avaloq's global CISO Governance team and define and monitor the company's Information Security Management System.
- Own security policy and governance direction.
- Provide second-line oversight of first-line security functions.
- Drive compliance monitoring and control effectiveness in a regulated fintech environment.

Amazon Web Services — *Switzerland*
Global Security Architect / Senior Global Security Architect — *Sep 2021 - Dec 2023*

Worked with global AWS customers and internal teams to improve security posture, define secure architecture patterns and scale security awareness.
- Led security architecture in the Stellantis Virtual Engineering Workbench program.
- Established security workstreams, metrics and reporting for business and technical stakeholders.
- Helped launch and scale Security Champions and Security Guardians initiatives.
- Supported secure collaboration mechanisms for the Volkswagen Digital Production Platform.

Swisscom — *Switzerland*
Enterprise Architect and IT Clouds Solution Security Architect — *Apr 2019 - Sep 2021*

Led security architecture, risk and compliance activities for Swisscom cloud platforms across multiple services.
- Directed compliance and governance activities including ISO27001, ISAE3402/3000 and GDPR requirements.
- Built and led a Security Champions community across IT Clouds teams.
- Coordinated threat modeling, audits, penetration testing and vulnerability management.

Team Lead & Product Owner for Health & State Management — *Aug 2014 - Apr 2019*

Built and led the Health and State Management team for Swisscom cloud platforms.

- Managed a team of up to 16 people.
- Owned roadmap and prioritization with product managers and stakeholders.
- Defined and delivered platform-wide monitoring, logging and alerting capabilities.
- Led the *Orchard* project through its implementation, production release and further improvements and development.

CFEngine AS
Norway/U.S.A. (remote)
PRODUCT MANAGER
Oct 2011 - Jun 2014

- CFEngine Advocate, with a special focus on security.
- Managed the CFEngine language roadmap.
- Created and led the CFEngine Design Center project, which was the foundation for the current CFEngine Build service.
- Coordinated the work on CFEngine third-party integration (e.g. AWS EC2, VMware, Docker and OpenStack).
- Wrote the book *Learning CFEngine 3*, published by O'Reilly Media, which became the de facto introductory text to CFEngine.

IBM Zurich Research Lab
Switzerland
RESEARCH STAFF MEMBER
Oct 2001 - Oct 2009

Member of IBM Zurich's Global Security Analysis Laboratory, working on intrusion detection, malware containment and virtualization security research.

National Autonomous University of Mexico (UNAM)
Mexico
FOUNDER AND LEAD OF COMPUTER SECURITY AREA
Aug 1995 - Aug 1996

- Established UNAM's *Computer Security Area*, the University's first team dedicated to computer security, which has evolved into the *Information Security Coordination (UNAM-CERT)*.
- Managed up to nine people working on different projects related to computer security.
- Provided security services to the whole University, including incident response, security information, auditing and teaching.

Education

Purdue University
West Lafayette, IN, U.S.A.
PH.D. IN COMPUTER SCIENCE
Aug 1996 - Aug 2001

- Thesis: Using Internal Sensors for Computer Intrusion Detection. Advisor: Eugene H. Spafford

Purdue University
West Lafayette, IN, U.S.A.
M.S. IN COMPUTER SCIENCE
Aug 1996 - May 1998

- Advisor: Eugene H. Spafford

National Autonomous University of Mexico (UNAM)
Mexico City, Mexico
BACHELOR'S DEGREE IN COMPUTER ENGINEERING
Aug 1989 - Jul 1995

- Thesis: UNAM/Cray Project for Security in the Unix Operating System (in Spanish, original title: *Proyecto UNAM/Cray de Seguridad en el Sistema Operativo Unix*)

Certifications

Full list available at Credly

Certified Information Systems Security Professional (CISSP)
Apr 2019
ISC2

Selected Research

See Publications for publication details.

Billy Goat: Active worm detection and capture
2002 - 2008
IBM RESEARCH

- Pioneered active worm-capture technology that became the foundation for modern honeypots and honeynets
- Designed system to simulate thousands of vulnerable hosts to attract and capture propagating worms
- Implemented automated analysis to extract signatures and update intrusion detection/prevention systems
- Publications: [18], [25]

Ph.D. Thesis: Using internal sensors and embedded detectors for intrusion detection
1996 - 2001
PURDUE UNIVERSITY

- Investigated novel approaches to data collection for intrusion detection systems
- Implemented and evaluated internal sensors and embedded detectors as data collection mechanisms
- Analyzed trade-offs and effectiveness of different sensor placement strategies in IDS architectures
- Publications: [11], [19], [20], [27], [26], [32]

Software

Open-source software projects

- GitHub: github.com/zzamboni
- GitLab: gitlab.com/zzamboni

Honors & Awards

May 2020	**IEEE Security & Privacy Test of Time Award (IEEE S&P page, CERIAS blog post)**, IEEE
2010	**CFEngine Champion**, CFEngine AS
May 1996	**Fulbright Scholarship (for pursuing Ph.D. studies at Purdue University)**, Fulbright Program and CONACYT

Selected publications

Books

[1] Diego Zamboni. *Publishing with Emacs, Org-mode and Leanpub*. Leanpub, June 2020.

[2] Diego Zamboni. *Learning CFEngine*. O'Reilly Media, Inc. 2012–2017, afterwards self-published, 2012.

Editorial Activities

[3] Diego Zamboni and Christopher Kruegel, editors. *Proceedings of the Recent Advances in Intrusion Detection (RAID): 9th International Symposium*. Lecture Notes in Computer Science. Secaucus, NJ, USA: Springer-Verlag New York, Inc., 2006.

Theses

[4] Diego Zamboni. "Using Internal Sensors for Computer Intrusion Detection". CERIAS TR 2001-42. PhD thesis, West Lafayette, IN: Purdue University, August 2001.

Refereed Papers

[5] James Riordan, Diego Zamboni, and Yann Duponchel. "Building and deploying Billy Goat, a worm-detection system". In: *Proceedings of the 18th Annual FIRST Conference*. June 2006.

[6] Florian Kerschbaum, Eugene H. Spafford, and Diego Zamboni. "Using internal sensors and embedded detectors for intrusion detection". In: *Journal of Computer Security* 10.1,2 (2002), pages 23–70.

[7] Christoph L. Schuba, Ivan V. Krsul, Markus G. Kuhn, Eugene H. Spafford, Aurobindo Sundaram, and Diego Zamboni. "Analysis of a Denial of Service Attack on TCP". In: *Proceedings of the 1997 IEEE Symposium on Security and Privacy*. Awarded the 2020 IEEE Security & Privacy Test of Time Award. IEEE Computer Society, IEEE Computer Society Press, May 1997. pages 208–223.

Skills

Leadership	32 years of multidisciplinary team and project leadership experience · IT Enterprise Architecture · Scaled Agile Framework (SAFe)
Information and Cyber Security	Enterprise security governance · Enterprise security architecture · Virtualization and cloud computing security · Risk management and compliance · Intrusion detection and prevention · Software security and secure software development · ISO27001
Technology	Broad and deep IT expertise · Cloud computing · Computer security · Operating systems · Networking · Configuration management · Software & services development · Programming languages
Research	Ph.D. in Computer Science · 9 years of experience at IBM Research
Communication	Excellent written and spoken communication skills · Extensive public speaking experience · Professional writing and teaching experience

Languages

Spanish	Native
English	Full proficiency
German	Intermediate proficiency (B2 level)

References

Available by request